ProteinNetworkSight Tutorial

Summary

Our tool works with two existing databases:

- Intact db, an online tool and database of known and predicted protein-protein interactions
- Cancer Dock Database, a database that provides a listing of licensed cancer drugs

ProteinNetworkSight is an open source tool for calculating patient-specific protein networks based on integrating the input from the user with the information retrieved from Bindx db.

Starting Data

For the user to begin, they must upload a Excel CSV (.CSV) file in the format shown in a TSV that represents an Excel file), where each row represents a protein in gene and each column represents a factual. The first row contains more than 5000 rows, where each row represents a proteingene and the columns are features to be analyzed. The first column contains a column of proteingene names as well as additional columns of numeric values. These numeric values represent gene/protein scores or quantitative metrics - such as leadfoldgene expression, or mutation frequency. The first row contains the column headers, including the first column that changes. The first line is the header, specifying column names. Note that the names of the numeric columns should start with the same prefix.

Here are an Excel example file (Retrieve from Sequencer.xl, ng) Precision Oncology, 2021, Supplementary Data 3. For example, here are 10 rows of the example file, where the gene column is named "Gene", and the numeric columns are named "Expression" and "Mutation". Note that the prefix of the numeric fields that operate related. The prefixes the weights of proteins for calculating a composite patient score, as determining the information thresholds analysis presented in Sequencer et al. However, this format is flexible, the "Gt" columns can be replaced by any other scores, coefficients, or fold changes, depending on the specifically method the user analyzes to find or expression patterns.

Gene	Expression	Mutation
ZNF804A	0.000000	0.000000	0.000000
TMEM59	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000
ZNF804A	0.000000	0.000000	0.000000

Setting parameters and thresholds

In this step the user should specify:

- The header name of the gene/protein gene column (in the example file it is "id1").
- The prefix shared by the names of the numerical columns (in the example file it is "Gt"). (i.e. a weight of each gene/protein for a patient-specific network).
- Thresholds for information theory (with the Sting interaction score, representing probability of interaction between each pair of weighted proteins. The thresholds are defined according to STRING db.
- Positive and negative thresholds for retrieving edges (based on the range of protein scores (for example what is the min/medium score to be included in the network analysis)
- Organism of interest (in the example file, it's "Homo sapiens")

Manual thresholds adjustments

- You can also choose to adjust the thresholds of every S column in the graph individually.

Adjusting proteingene names

Our tool connects to Bindx db, in order to create and visualize protein networks. Thus, proteingene names specified in the input file should match the names used by String-db. Genes often have multiple names, so it is important to make sure that the names are in a format that Bindx db can recognize. Note the short String-db. The tool may also find high hub-score genes and allow the user to rename them.

Hub Genes Table
Gt1 Gt2 Gt3 Gt4 Gt5 Gt6 Gt7 Gt8 Gt9 Gt10